From specimen to sequence: Huang MEGA Lab Nanopore workflow

amplicon (two-step or one-pot PCR) or whole genome · scan or click a QR code to open the protocol · LIMS stages 1 → 2 → 3a → 3b → 4a

Stage 1
Specimen# / Culture#

Specimen
soil / environmental / insect | fungal strain in culture

`/lims-register`
Stage 1 → Specimen# (soil, environmental) or Culture# (strain)

isolate fungi (strains, Route B / C)

soil / environmental sample,
or an existing pure culture:
extract directly (skip isolation)

Stage 2
isolate + culture · Culture#

Microbe isolation
environmental sample → vial

Culture media
MEA, PDA and other recipes

Isolation and culturing
primary plate on MEA (Stage 2a) → subculture one colony / hyphal tip on PDA
one Stage-2b row per isolate · preserve in vials
sequence from a pure culture only

`/lims-register`
Stage 2a plate → Stage 2b isolate → Culture# per strain

DNA extraction

Stage 3a
extraction + Qubit · MolWork#

Soil DNA extraction
SuperSoil · Zixpress 32

Fungal DNA extraction
modified BOMB v9.0

DNA extraction
soil: SuperSoil, Zixpress 32 | fungal / HMW gDNA: BOMB v9
genome route: keep the DNA high-molecular-weight

quantify

Qubit Flex
concentration of the extract (dsDNA HS)

`/lims-register`
Stage 3a → MolWork# (minted only after quantification)

choose a route

Stage 3b
PCR + gel + Qubit · pcr_reaction

Route A · amplicon, two-step PCR
environmental samples (soil, water) and mixed templates

1st PCR (gene regions)

2-step PCR protocol
mixtures + cycling

1st PCR products (per sample)
🟦 head seq — gene region —

2nd PCR (barcoding)
gel check #1: good amplicons only

2nd PCR products: one tube per sample, one barcode per tube

Tube 1 ▶ barcode 1
Tube 2 ▶ barcode 2
• • •
Tube n ▶ barcode n

▶ barcode 🟦 head seq — gene — 🟦 head seq ◀ barcode

Route B · amplicon, one-pot PCR
strains: single- or multi-locus

Set up one-pot PCR

One reaction tube (per strain × locus)
gene-region primers
(head-tailed, 0.1 µM each)
barcode primers
(barcode + head seq, 0.3 µM each)
50 µL: KAPA HiFi 2X + both primer sets + 3 µL template

One-pot PCR protocol
50 µL mix + 25/15 cycles

One-pot PCR
phase 1: 25 cycles, 60–66 °C
phase 2: 15 cycles, 65 °C

One-pot PCR products: one tube per sample, one barcode per tube

Tube 1 ▶ barcode 1
Tube 2 ▶ barcode 2
• • •
Tube n ▶ barcode n

▶ barcode 🟦 head seq — gene — 🟦 head seq ◀ barcode

same barcoded molecule from either route

Gel check → Qubit Flex (dsDNA BR)
confirm the barcoded product

`/lims-register`
Stage 3b → pcr_reaction: method, conc., 2nd_pcr_ok, barcode

Pooling (equal molar)

Pool all barcoded samples
equal molar of every sample

Purification (bead clean-up)
PCR sample : bead = 1 : 1

Size selection 1X
bead purification protocol

Library construction

KAPA EvoPrep library
PCR-barcoded pool, ~200 fmol
end-prep → adapter ligation → clean-up
PromethION

EvoPrep library prep
KAPA EvoPrep NBD114.96

Route C ↓

Route C · genome, no PCR (Stage 3a → 4a, no 3b)

QC the gDNA

High-molecular-weight gDNA
integrity: gel / TapeStation
input per ONT table
(10 kb avg → 200 fmol ≈ 1.3 µg)

Library construction

KAPA EvoPrep + NBD114.96
end-prep → native barcode
ligation (NB01–96) → pool
→ adapter ligation
~140 min

EvoPrep library prep
KAPA EvoPrep NBD114.96

Stage 4a
Nanopore experiment · Nanopore#

`/lims-register`
Stage 4a → Nanopore#: experiment, samples (PCR barcode or NB), flow cell, NanoAmp

Load the flow cell and sequence
PromethION R10.4.1 (FLO-PRO114M)

run data (POD5)

`/nanoamp`
amplicon runs (Routes A and B)
basecall → demux → trim → cluster → BLAST
run it on the run folder with the LIMS sample sheet

`/nanogenome`
genome runs (Route C)
basecall → assemble → annotate → compare
run it on the project directory

after the run
(bench)

Flow cell wash
EXP-WSH004 PromethION

Flow cell wash and reuse
EXP-WSH004

`/lims-register`
Stage 4a → flow cell: pores after wash, date of use

Legend

- 🟦 Head seq: universal tail on the gene-region primers
- ▶ Barcode: sample-specific, on the barcode primers
- — Gene region: target amplicon (colour = sample)

QR code: scan, or click on the website, to open that step's protocol PDF
Left bands = LIMS stages; /lims-register ends every stage; dashed = QC / info
Route C uses native barcodes (NBD114.96) in the library: no PCR, no Stage 3b
/nanoamp, /nanogenome: Claude Code analysis skills for the run data

Huang MEGA Lab (NCKU) · chart built 2026-10-21 from doc/protocols/registry.csv · live version with clickable links: ythuangmyco.github.io/protocol